

Priyanka Mondal

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🐙 Priyanka-Mondal • in mondalp

Summary

- 4 years of experience as **Software Engineer** and 6+ years of experience as **Security & Privacy Researcher**
- **Interests:** Applied Cryptography, Security and Privacy in Distributed Systems, Data Governance, Threat Modeling

Skills

Languages: C++, Python, Go, Java, SQL, Javascript, Haskell, Coq, HTML/CSS, Matlab

Tools & Platforms: Git, GCP, AWS, Docker, Kubernetes, OpenSSL, Linux/Unix, VS Code, Cursor

Education

Ph.D., Computer Science , University of California, Santa Cruz	2017-24
Master of Engineering, Computer Science , Indian Institute of Science, Bangalore	2013-15
Bachelor of Engineering, Computer Science , Bengal Engineering & Science University, Kolkata	2009-13

Experience

- **Snap Inc., Santa Monica, CA | Privacy-by-Design Team** *Software Engineer L4*, Sept 2024 – Apr 2026
 - **Privacy Incident Response and Data Governance**
 - Investigated privacy incidents, identified root causes, proposed mitigations, and collaborated with 15+ feature teams to drive timely resolution and long-term fixes.
 - Built an encrypted data classifier and supported data mapping efforts to improve visibility into data sensitivity and protection requirements.
 - **Development of Privacy Validation Tools**
 - Built a daily validation pipeline to detect violations of user location-sharing preferences based on production logs.
 - Developed dashboards and alerting system to continuously monitor privacy regressions.
 - **Privacy Audit Pipeline for Machine Learning Models**
 - Applied **membership inference attacks** to empirically measure privacy leakage and validate differential privacy guarantees for ads ranking models.
 - Built an AI agent to trigger privacy audit workflows for production ads ranking models, and monitor for failures and audit requirement violations.
 - **Privacy Reviews and Privacy Analysis**
 - Conducted privacy analysis and 130+ privacy reviews for features across Ads Ranking, Content Ranking, Camera Roll, and Friend Graph, to assess privacy risks and compliance requirements.
 - Designed privacy-enhancing approaches using differential privacy and cryptography for sensitive product use cases including Face Tagging, Ads targeting, and Friend suggestions ranking.
- **University of California, Santa Cruz | Security Research Lab** *Graduate Research Assistant*, 2017-24
 - **Secure and Efficient search on Encrypted databases ensuring Forward and Backward Privacy**
 - Designed and implemented an encrypted search algorithm that improves database search time more than **100×** on HDDs and SSDs than the existing counterparts.
 - Implemented a secure data-structure (Oblivious RAM) using cryptographic mechanisms and B-trees, reducing the access time by **2-6×** than the existing AVL-tree based construction.
 - **FLAQR: A programming model to securely implement Consensus, Replication and Secret-sharing**
 - Designed a programming model to write fault-tolerant distributed applications that are secure-by-

construction.

- Worked on Type Systems, Formal Verification, Haskell Libraries and Information Flow Control Policies.

- **Citrix R&D Pvt. Ltd, Bangalore. | Networking & Cloud Team** *Software Engineer II, 2015-17*
 - Implemented an algorithm in **Python** to transmit **JSON** data from Packet Engines to Amazon S3 buckets.
 - Contributed to the reliability of the NetScaler load balancer by investigating and fixing 20+ software issues.
 - Developed an **Wireshark** plugin that increased efficiency of **HTTP/TCP** packet testing by **30%**
- **Nomura Research Institute, Kolkata. | Enterprise Data Warehouse Team** *Summer Intern, 2012*
 - Deployed an automated **Java** parser for incoming **XML** data packets, improving system throughput by **70%**.

First Author Publications

1. P. Mondal, J. G. Chamani, I. Demertzis, and D Papadopoulos. *I/O-Efficient Dynamic Searchable Encryption meets Forward & Backward Privacy*. **33rd USENIX Security, 2024**
2. P. Mondal, M. Algehed and O. Arden. *Flow-Limited authorization for consensus, replication, and secret sharing*. **31st Journal of Computer Security, 2023**
3. P. Mondal, M. Algehed and O. Arden. *Applying consensus and replication securely with FLAQR*. **35th IEEE Computer Security Foundations, 2022** ([Distinguished Paper Award](#))